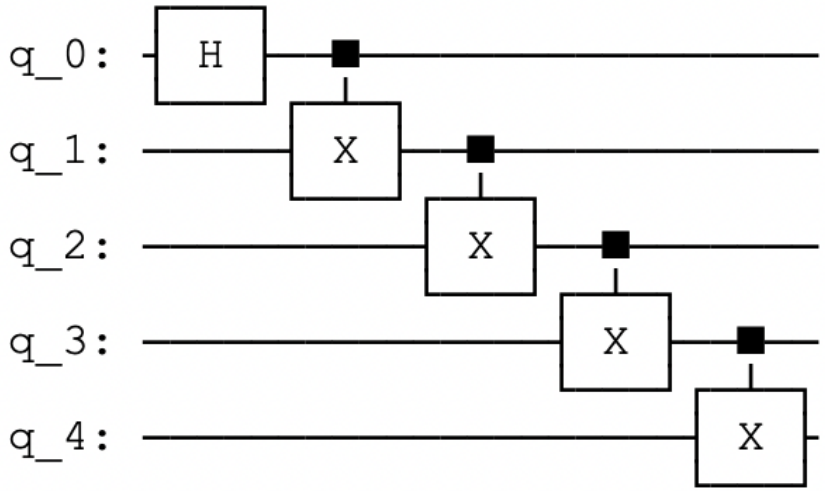
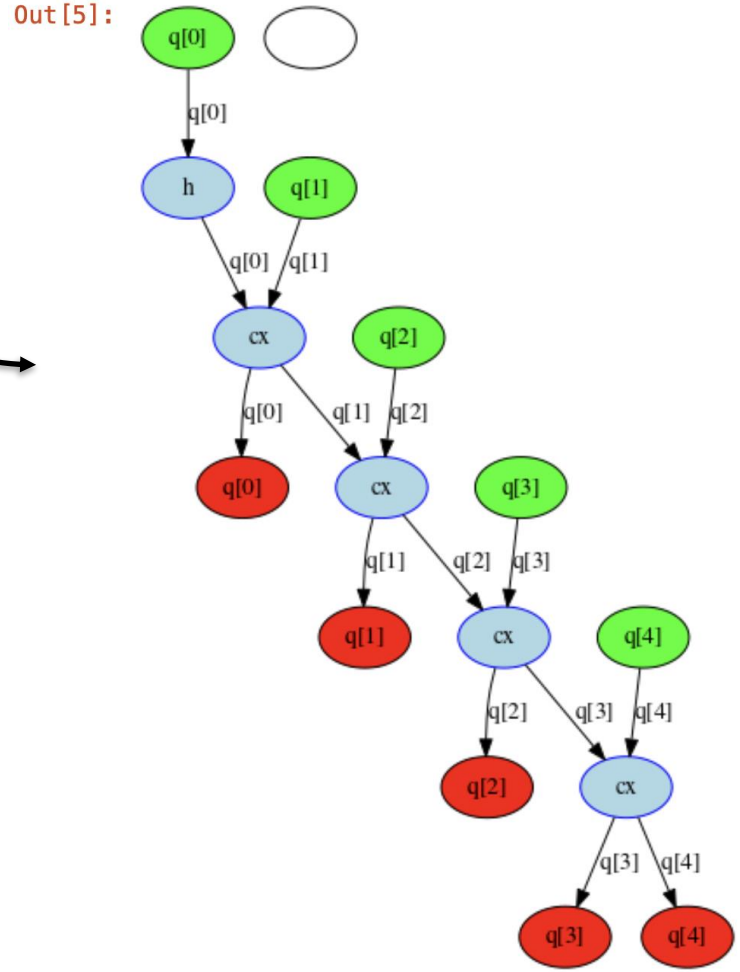
Quantum-Sparse Numerical Library

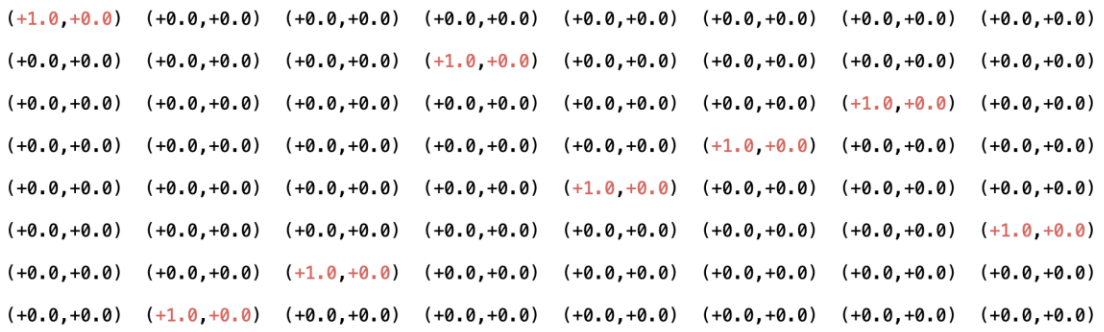
Blake Burgstahler
Srikar Chundury

How does simulation work internally?

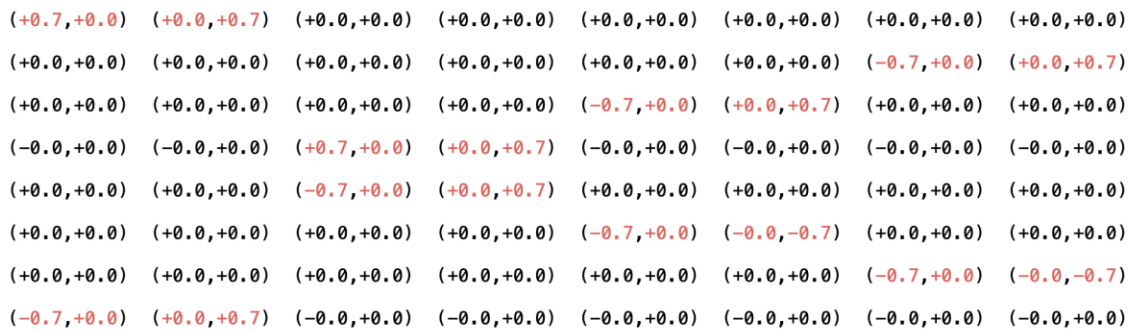




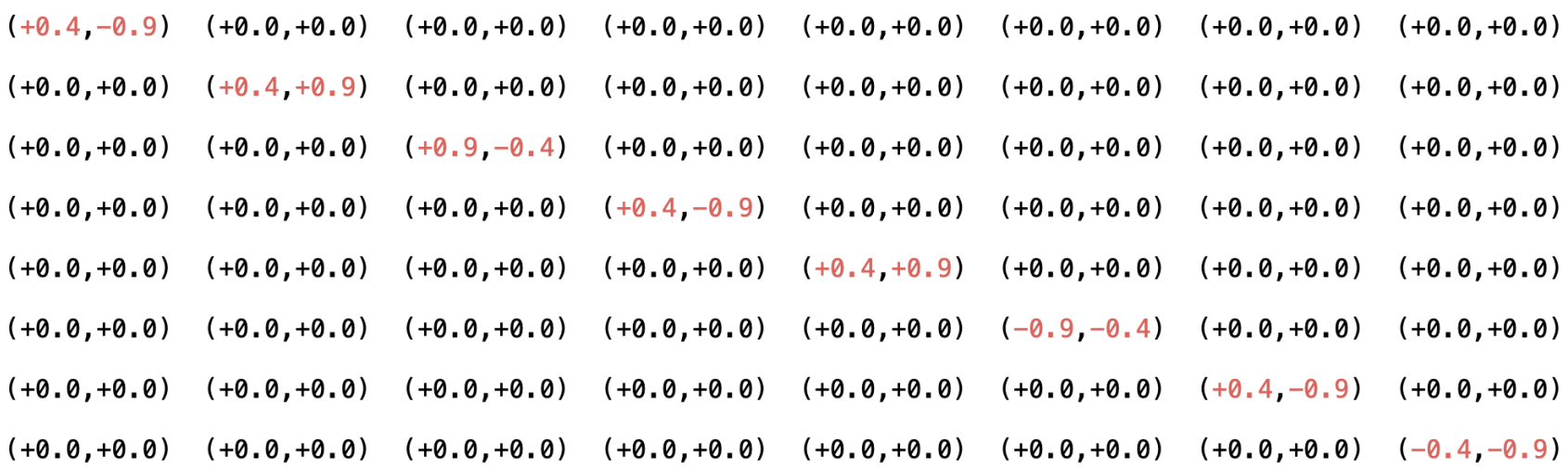
Analyzing Quantum Sparsity



Supermarq's GHZ circuit



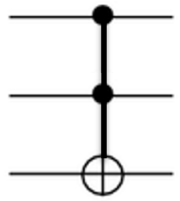
Supermarq's Mermin Bell circuit



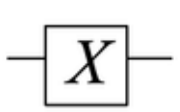
Supermarq's Hamiltonian circuit

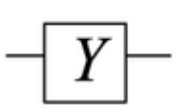
Simplifying Assumptions

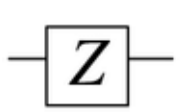
- ”Tensors” :: Matrices
- Tend to be very sparse
- Non-zeros tend to be +/- 1 (potentially scaled)
- Non-zeros tend to fall on diagonals

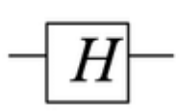


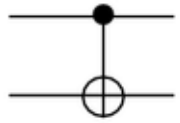
$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$











$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Focus on Diagonals

$$S = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}; \quad A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 & 1 \\ 1 & 2 & 3 & 2 & 1 \\ 1 & 2 & 2 & 2 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

Then,

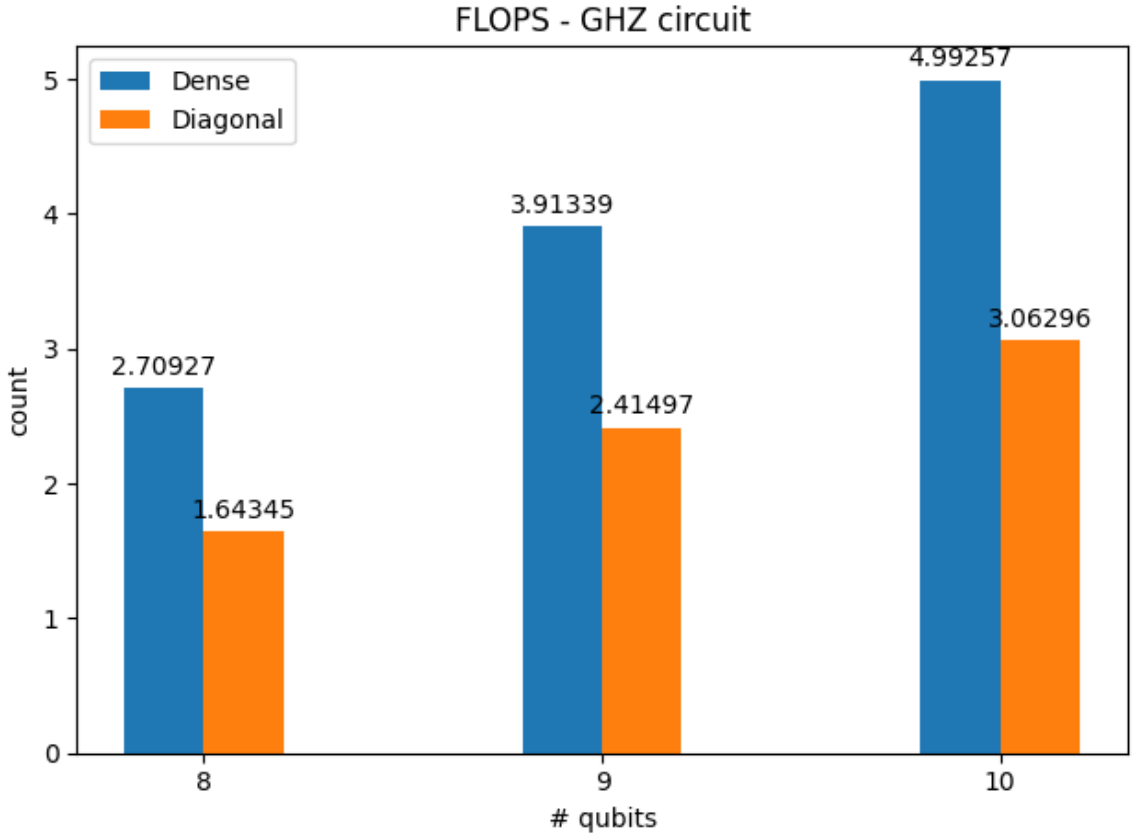
$$SA = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 2 & 1 \\ 1 & 2 & 3 & 2 & 1 \\ 1 & 2 & 2 & 2 & 1 \end{pmatrix}; \quad AS = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 1 & 0 \\ 2 & 3 & 2 & 1 & 0 \\ 2 & 2 & 2 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

- Matrix Product :: Only requires diagonals
- Product of diagonals: O(n)
  - Each of which is a shift and elementwise multiplication
- Overall: $O(d_1 * d_2 * n)$
- ‘Sparse Diagonals’ and ‘Compressed Sparse Diagonals’

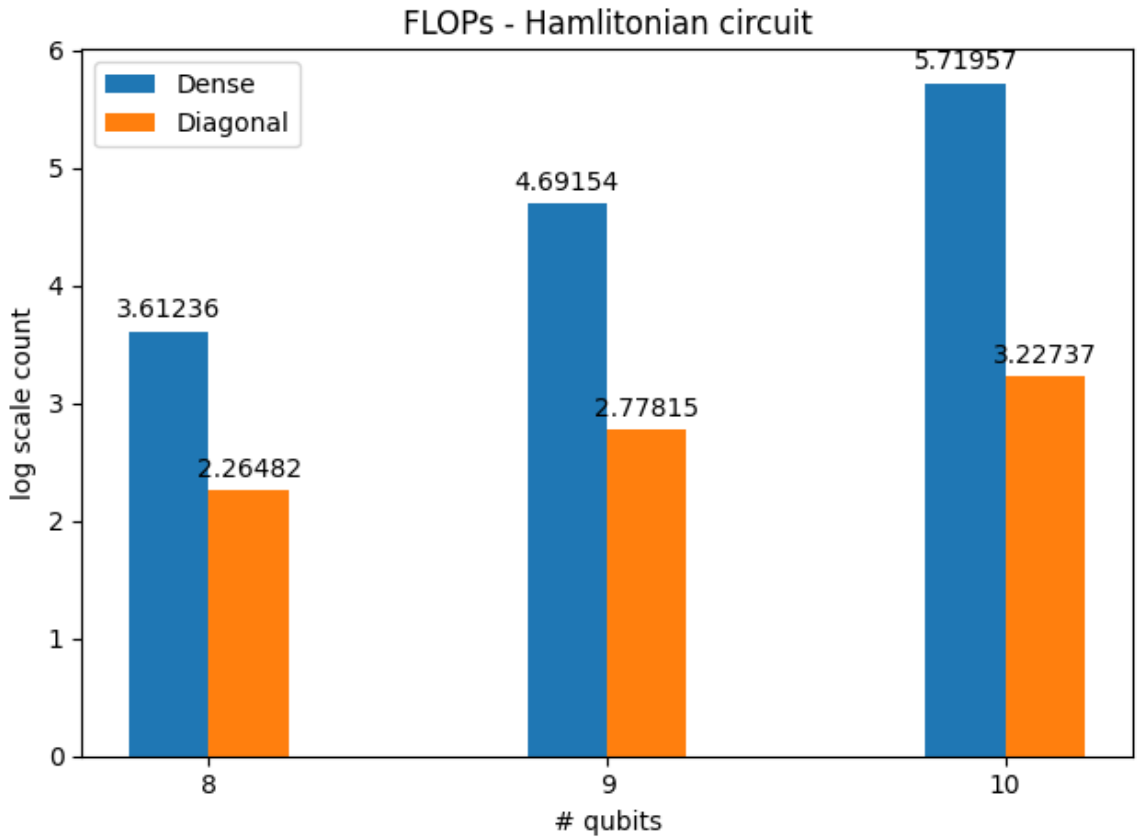
Experiments

- SupermarQ Benchmark: 10 qubit circuit ⟶1024x1024 matrix
  - GHZ (**9** matrix multiplications)
  - Mermin Bell (**78** matrix multiplications)
  - Hamiltonian (**37** matrix multiplications)
- Exported the intermediate matrices (gate unitaries) using Qiskit to .npz files.
- Imported them and read them into our data structure and ran multiplications serially.
- TODO: Run of ARC for larger #qubits and randomly generated matrices. Implement "Compressed Sparse Diagonal" version.

Sample Results



Supermarq's GHZ circuit



Supermarq's Hamiltonian circuit

Questions?

